TOH YUN NING

Data Science Graduate — Machine Learning, Data Engineering & Semiconductor Manufacturing Analytics
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EDUCATION

Tunku Abdul Rahman University of Management and Technology (TAR UMT), Penang — Jul 2024 – Jul 2026
Bachelor of Data Science (Honours)
CGPA **3.8** with President's List (4.00) & Dean's List (3 trimesters), held on a full merit scholarship. Focused on Data Analysis & Computer Vision

Tunku Abdul Rahman University of Management and Technology (TAR UMT), Penang — Jul 2022 – Jun 2024
Diploma in Computer Science
CGPA **3.9**, focused on programming fundamentals, data science basics and statistics

INTERNSHIP EXPERIENCE

Gospower Electric Technology Co., Ltd., Dongguan, China — Jan 2026 – Jul 2026
Research and Development Department Intern

- Rotated across a server power-supply (CRPS) manufacturing pipeline — production, SMT/AOI, EMC/reliability testing, automation, and QC — gaining ground-level exposure to manufacturing defect-tracking systems.
- Ran automated test campaigns on a Chroma 8000 system (11 test types incl. surge current, OCP, dynamic load).
- Gained direct exposure to yield/defect-tracking pipelines in high-volume electronics manufacturing.

ViTrox Technologies Sdn Bhd, Batu Kawan, Penang, Malaysia — Oct 2023 – Feb 2024
Research and Development Department Intern

- Labelled and trained PyTorch-based image segmentation on PCB defect datasets to support automated X-ray inspection (AXI) R&D.
- Produced technical evaluation reports benchmarking model performance.

FINAL YEAR PROJECT

Void Segmentation in Land Grid Array (LGA) Components Using DeepLabV3+ and SegNeXt
with data sponsored by ViTrox Technologies Sdn Bhd

- Refined and compared SegNeXt and DeepLabV3+ for detecting solder voids in X-ray images of LGA components (111 images, 873 joints).
- Extended the project after submission by rebuilding the evaluation as five-fold cross-validation split by source image, removing test leakage from the original split: 85.8% mean IoU, with SegNeXt matching DeepLabV3+ at one-fifth the parameters (6.0M vs 28.6M).

NOTABLE ASSIGNMENTS

Local PDF Question-Answering Chatbot (RAG) — Natural Language Processing

- Owned the retrieval and generation pipeline: document chunking, embeddings (nomic-embed-text), ChromaDB vector store, MMR retrieval and LLaMA 3 served locally over Ollama's REST API; also contributed to the web application.

Malay Lexicon Data Pipeline — Data Engineering

- Built the ingestion layer: web scrapers and data cleaning for dictionary and proverb sources, plus the Kafka producer and consumer that stream data into Hadoop and PySpark for processing.
- The pipeline stores the lexicon in Neo4j with Redis caching and refreshes on scheduled cron runs.

Bus Station Management System — Advanced Database Management

- Built a multi-table Oracle schema with compound triggers for cascading business rules and audit logging, and stored procedures with exception handling for bulk booking, seat validation and pricing.

TECHNICAL SKILLS

Languages: Python, R, Java, C, JavaScript, HTML/CSS, SAS
ML/AI: PyTorch, scikit-learn, BERT/Transformers, computer vision (segmentation, watermarking), NLP (Llama)
Data Engineering: Kafka, Spark, Hadoop, ETL, data warehousing (star/constellation schema)
Databases & Tools: Oracle, MySQL, Neo4j, Redis, Power BI, SAS Viya, Git, Linux, Jupyter, Google Colab